
ECS 152A Computer Networks
Instructor: Liu

Name:

Student ID #:

Final Exam: March 17, 2005

Duration: 120 Minutes

1. The exam is closed book. However, you may refer to one sheet of A4 paper (double sided) of your own notes.
2. Try to solve as many as can.
3. Be brief.
4. Show your reasoning clearly. If your reasoning is correct, but the final answer is wrong, you will receive most of the credit.
5. Write your solution on this paper. If you need extra paper, use plain white paper. Write your name on any extra paper.

Good Luck!

Do not write in this box.

Question	1	2	3	4	5	6	7	8	Total
Points									

1. True or false (points).

- 1) Loss of packets and/or acks in stop-and-wait ARQ necessitates timeouts in the design.
- 2) All DNS queries which are not found in the local cache go to the root servers.
- 3) Reliability is a necessary function to be provided by transport layer protocols.
- 4) A short link has a low propagation to transmission time ratio and therefore, even a stop-and-wait flow control would be efficient on it: there is no need for complexity of larger windows.
- 5) Consider a point-to-point link with no multiple-access, bit-errors or flow-control issues. Framing is not required for such a link.
- 6) Randomness (in service and arrival) is what causes queuing at buffers.
- 7) “Full-duplex” “Point-to-point” Gigabit Ethernet implies that only the Ethernet framing format is used, but the CSMA/CD protocol is not used (because CSMA/CD assumes a “half-duplex” and “multiple-access” channel).

2. (points) Filling the blanks.

	ARP	Internet Routers	DHCP server
Plug and play (yes or no)			
layer (one of the five layers)			

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3. (6 points) FDMA. Suppose an operator is allocated 12.5MHz of spectrum. A guard band of 10KHz is needed at the edge of the allocated spectrum band. Each voice channel requires 30KHz of bandwidth. What is the number of channels the operator can have using FDMA?

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4. (10 points) ARQ. Consider a 1 Mbps WAN channel with 10 msec propagation delay. Data packets are 1000 bits long while ACK/NAK packets are negligible in length. Window size $(N) = 5$ packets, and there are 10 packets to be transmitted. Assume a Selective Repeat ARQ protocol. Assume that the timeout window is set to be 0.1sec. Now assume the 10th packets which cross the forward channel are lost. Other packets and ACKs/NAKs are not lost or corrupted. How much time is required to complete the transfer of the 10 pkts and receive the final ack.

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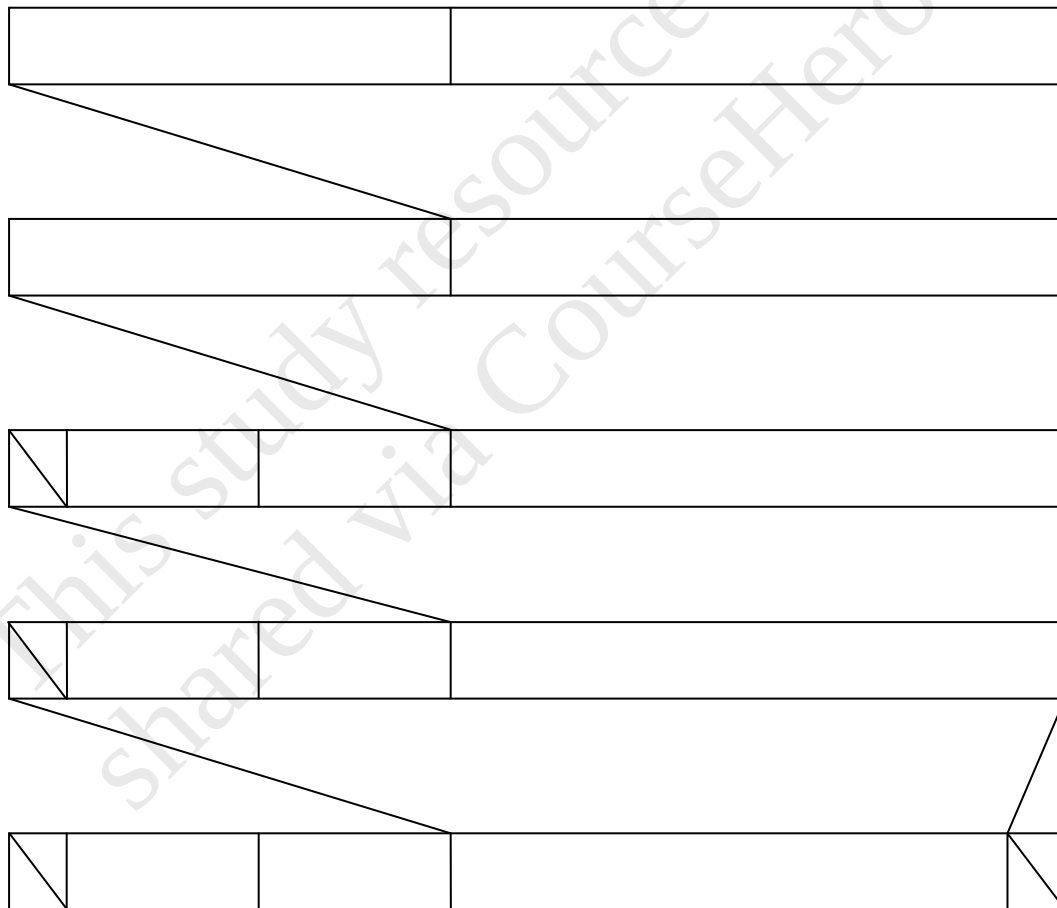
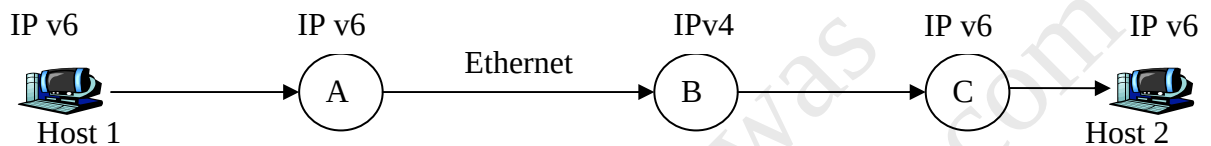
5. (10 points) Filling the blanks using the following terms: a modem, a telephone network, Ethernet, AM radio, a codec, analog transmission, digital transmission, digital signaling, analog signaling, physical layer, data link layer, network layer, transport layer, application layer, ADSL, cable, TDMA, FDMA, connection-oriented, connection-less, circuit-switching, and packet-switching. (Note that there are more terms than blanks.)

- 1) Among the five layers, _____ and _____ are implemented on end hosts but not on routers.
- 2) The Internet provides its applications two types of services. They are _____ and _____ services.
- 3) FM radio stations use _____ to separate different stations.

6. (10 points) Consider the following scenario. A student brings a laptop into CS building and uses the wireless LAN in CS building. She reads news from cnn.com. Name the networking protocols that the laptop is involved in and describe their functionalities briefly. You only need to specify protocols we discussed in class. You get full credit if you name and describe at least 6 correctly. (Note: irrelevant protocols will get negative credits.)

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7. (10 points) Encapsulation. A short email message is sent through the Internet. The link between routers A and B is an Ethernet link. Hosts and Routers A and C are IPv6 enabled, but Route B uses IPv4 only. So tunneling is used. The structure of the email message that is encapsulated in the Ethernet frame sent from Route A to B is drawn. Fill in the blanks: host1's MAC address, A's MAC address, B's MAC address, C's MAC address, host2's MAC address, host1's IPv6 address, A's IPv6 address, B's IPv6 address, C's IPv6 address, host2's IPv6 address, A's IPv4 address, B's IPv4 address, C's IPv4 address, IP packet, IPv4 payload, IPv6 payload, TCP header, UDP header, TCP payload, UDP payload, SMTP header, DHCP header, email message. Note that there are more options than blanks. For all addresses, assume source addresses is in front of destination addresses.



Extra questions:

Chpt. 3, p. 37

Chpt. 4, p. 9, p. 24

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